



PowerUp 2026

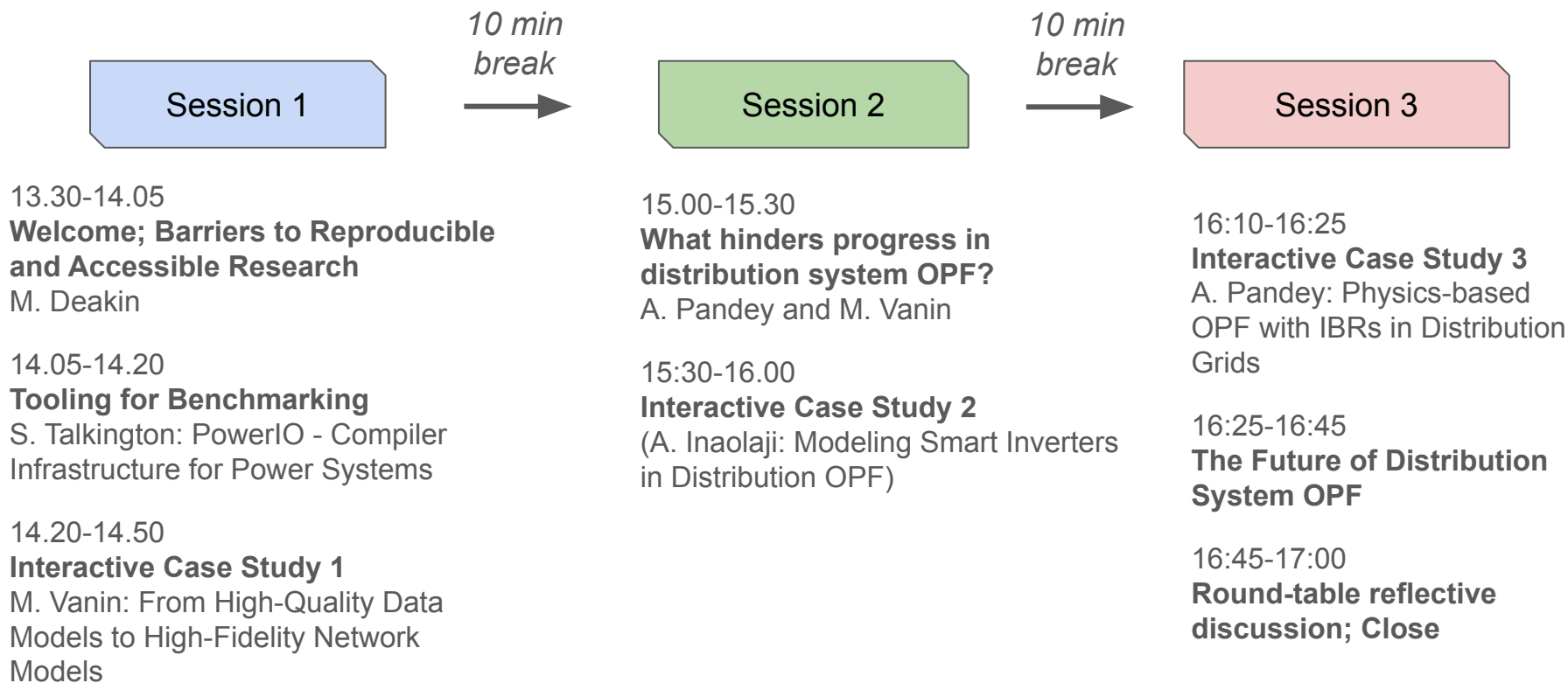
Towards a New Era of Reproducible, Transparent & Accessible Distribution System OPF Research

Friday 11th September 2026

Sponsor: IEEE Task Force on Benchmarking Multiconductor OPF for Distribution Systems



Agenda



Organisers and speakers



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**Samuel
Talkington**
Harvard SEAS /
Michigan EECS
samtalki

Preamble

- This workshop was developed to start conversations - please share comments freely (see also PowerUp Community Foundations)
- Feel free to post about how great is the workshop on linkedin / X / bluesky etc...
- ...but, please do not attribute comments to any particular person or organization
 - I.e., “Chatham House Rules”
- Please aim for minimum one person per table as Miro Board driver (next slide)
- Please join the Teams link (see emailed invite)

Workshop Objectives

1. Provide practical skills for attendees to support collaboration for reproducible and accessible research
2. Introduce and demonstrate via tutorials the BMOPF Task Force's work to-date, and provide a forum to discuss its ongoing & future directions
3. Provide an opportunity to experience (via interactive case studies) the data and models, and using them to solve OPF problems
4. Provide an opportunity for researchers, academics and industry interested in distribution system analysis to meet, exchange ideas, and form new connections and collaborations

Provocation: what we would like to answer this afternoon

1. How can the distribution system OPF community be encouraged to work on industrially relevant problems?
2. Is there a role for physics-based OPF for distribution systems in 10-20 years?
3. Are the main problems of *your* utility today addressable through an OPF, and could they form interesting benchmark problems?
4. What are the reasons that you do / do not make your research Reproducible, Accessible and/or Transparent, and, does AI change the calculus?
5. Should Reproducible, Accessible, and Transparent outputs be necessary for publication at PowerUp (and, longer-term, IEEE PES transactions, etc)?

-> comments can be added to the Miro Board, then, discussion in last session

Miro Board

1. How can the distribution system OPF community be encouraged to work on industrially relevant problems?

put your postit with an opinion / viewpoint here!

2. Is there a role for physics-based OPF for distribution systems in 10-20 years?

SETUP.md or Icebreaker

PowerUp 2026 attendees are all required to represent their country at the 2028 Olympics Games.

Discuss between you:

Which Olympic Sport would you least like to represent your country at?

...or, have a go to complete setup.md (see Teams chat)



<https://img-s-msn-com.akamaized.net/tenant/amp/entityid/AA1orVno.img?w=800&h=435&q=60&m=2&f=jpg>



<https://eu.usatoday.com/story/sports/olympics/2021/07/29/stunning-photos-synchronized-diving-tokyo-olympics/5404403001/>



<https://external-content.duckduckgo.com/iu/?u=https%3A%2F%2Ftse3.mm.bing.net%2Fth%2Fid%2FOIF.ljtSQWNPfZ9driW4J5DUwQ%3Fr%3D0%26pid%3DApi&f=1&ipt=76753985fe859bf013a70311565bbcb0aa0def3fc15ede836aca0de0256c20d8&ipo=images>

Barriers to Reproducible, Transparent and Accessible Distribution System OPF Research

Matt Deakin

Points

- Motivation: the **OPF-Native Distribution Utility** and what could they achieve
- **Distribution System OPF Sludge** that exists today that prevents uptake
- How **community benchmarking** is necessary for progress
- How **IEEE BMOPF Task Force will enable** community benchmarking
- A **Distribution System OPF Toolkit** for researchers / developers to enable reproducible, transparent and accessible outputs

Why Distribution System OPF?

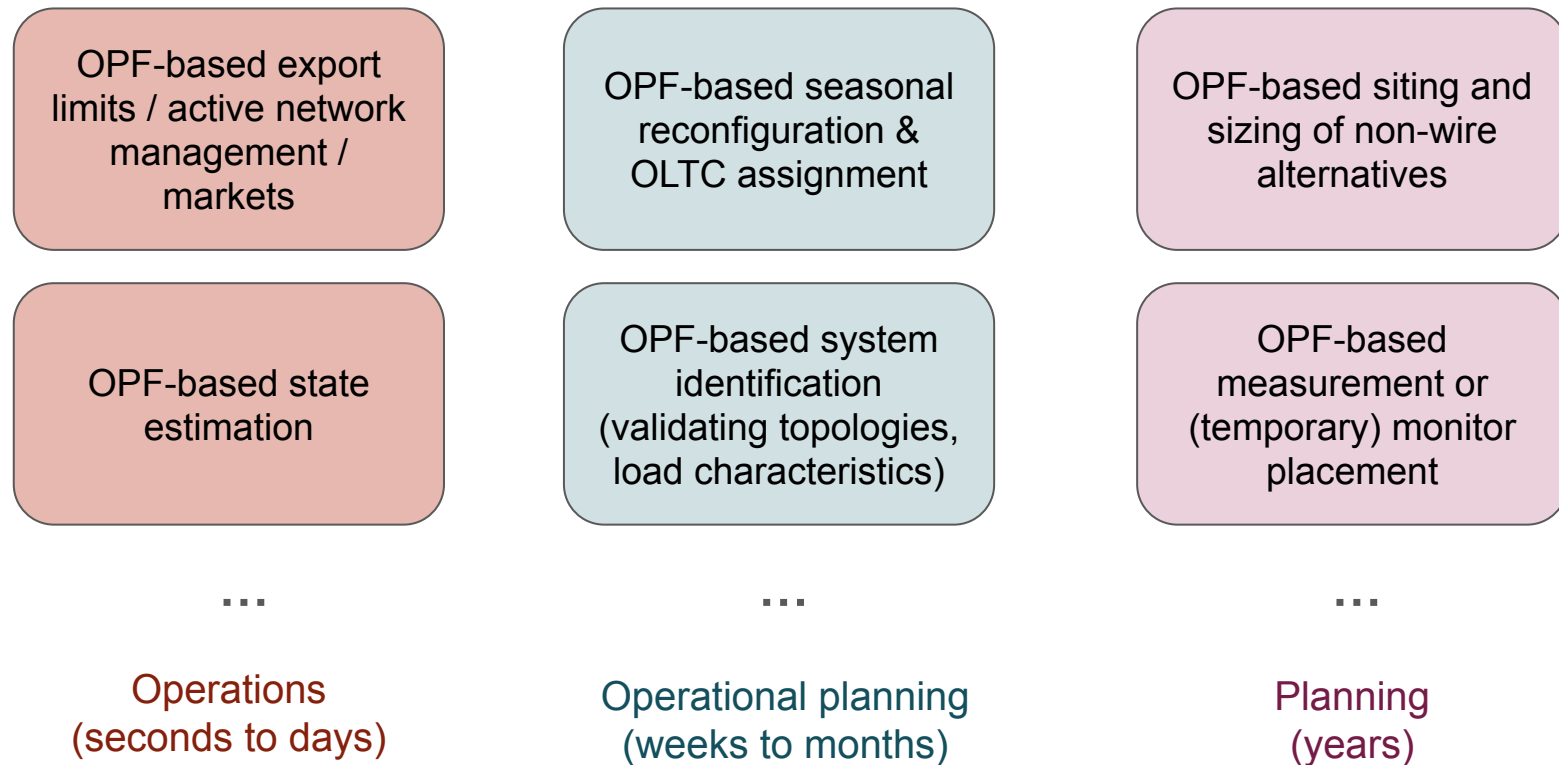
*Structural power system revolution in the **next 5-10 years**:*

- Large power stations, 100s MW -> Distributed energy resources
- Passive consumers -> Active prosumers
- Machine-based -> Inverter-based

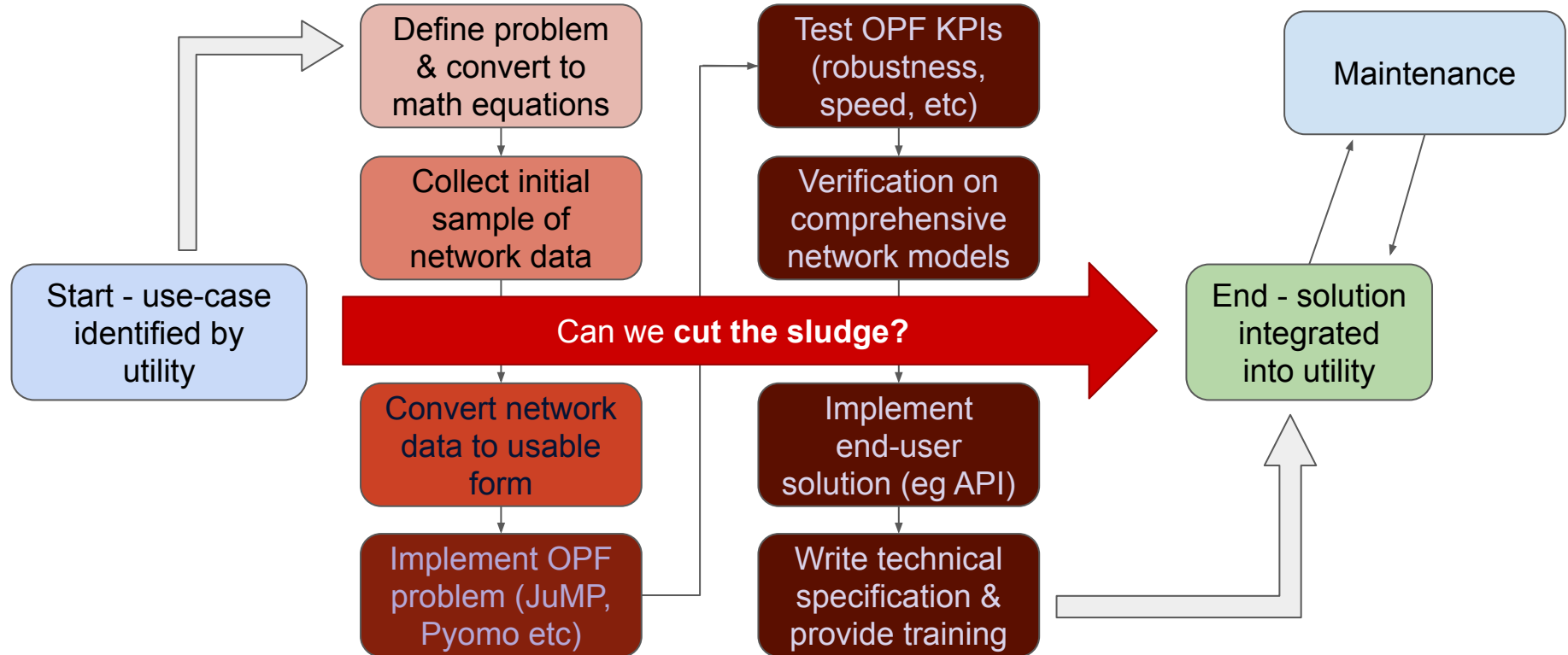
Utilities are feeling the heat

- Where can static export limits be relaxed safely for distributed energy resources?
- Can new measurements be used to address poor network models?
- What policies should be used to coordinate multitudes of domestic customer devices?
- How should actively controlled utility-owned assets be operated?

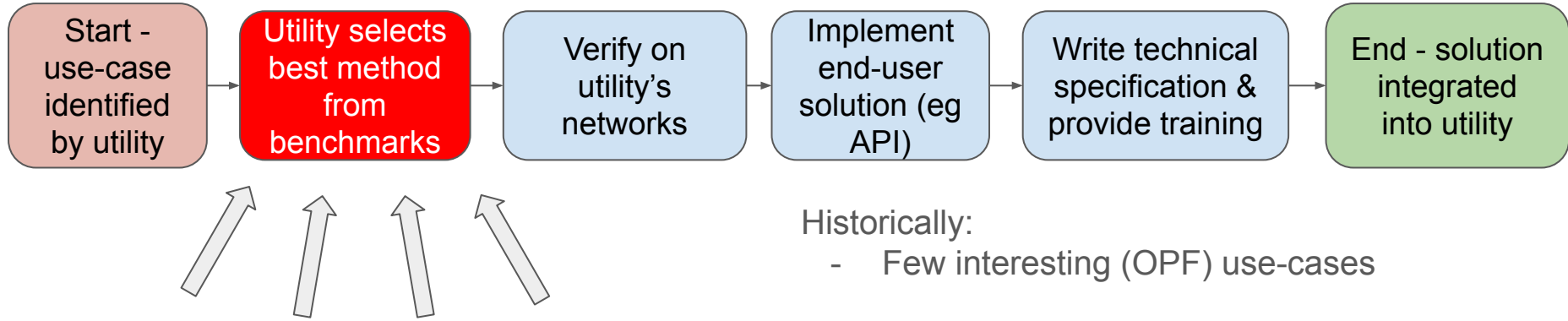
Vision of The OPF-Native Distribution System Operator: use-cases span operations to planning



What is required for optimization (OPF) problems to become used by distribution grid utilities?



What is required for optimization (OPF) problems to become used by distribution grid utilities?



Why don't we have this today?

As a **engineering research community** we could make progress on these through consensus →

Historically:

- Few interesting (OPF) use-cases

Today:

- Use cases slowly emerging ?
- Digitalization ?
- Data ?
- Smaller budgets vs TSOs ?
- **Physics ?**
- **Too much network variety ?**
- **Horrible data formats ?**

What could we agree as community to enable benchmarking?

- An appropriate level of physical abstraction (the ‘foundational physics’)
- A data format for defining OPF cases that is
 - Community accepted
 - AI-ready
 - Extendable for new / emerging use-cases and components
- A library of OPF cases to be used as benchmarks

Convened through “IEEE TF on Benchmarking Multiconductor OPF for Distribution Systems” (BMOPF TF) maintaining:

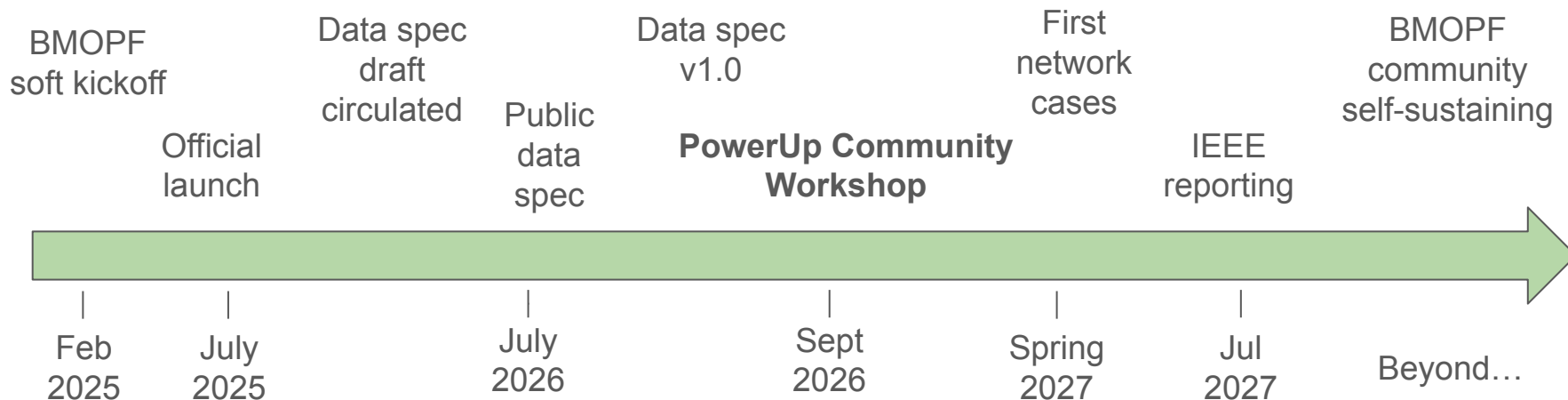
1. Data specification
2. Network cases

What is the status of the Task Force ?

- Open membership (today: 100+ mailing list)
- 9 data specification maintainers; 30+ Task Force member reviews
- Explicit governance for contributors going forward



[Join mailing list](#)



Home

- New here? Start with the Overview
- Foundations
- Components
- Optimisation
- Reference
- Contributing

Foundations

- Overview
- Background & scope
- Notation
- Data input formatting

Version

dev

Distribution System Model and Data Specification

Welcome to the data specification of the four-wire distribution-system optimal power flow (OPF) model.

This site is the canonical, version-controlled home of the specification: every page is plain Markdown that evolves through pull requests.

Download as PDF

The complete specification is also available as a single, print-ready document:
[DistributionSystemModelSpecification.pdf](#).

The data specification is one of two outputs maintained by the IEEE PES Task Force on Benchmark Multiconductor OPF (BMOPF) for Distribution Systems:

- The **BMOPF data specification**. This defines the basic specification which must be met for a dataset to be considered a network case.
- The **BMOPF network cases**. These are data instances which conform to the BMOPF data specification. Network cases are presently work-in-progress.

An JSON file representing a distribution system which meets the data specification is described



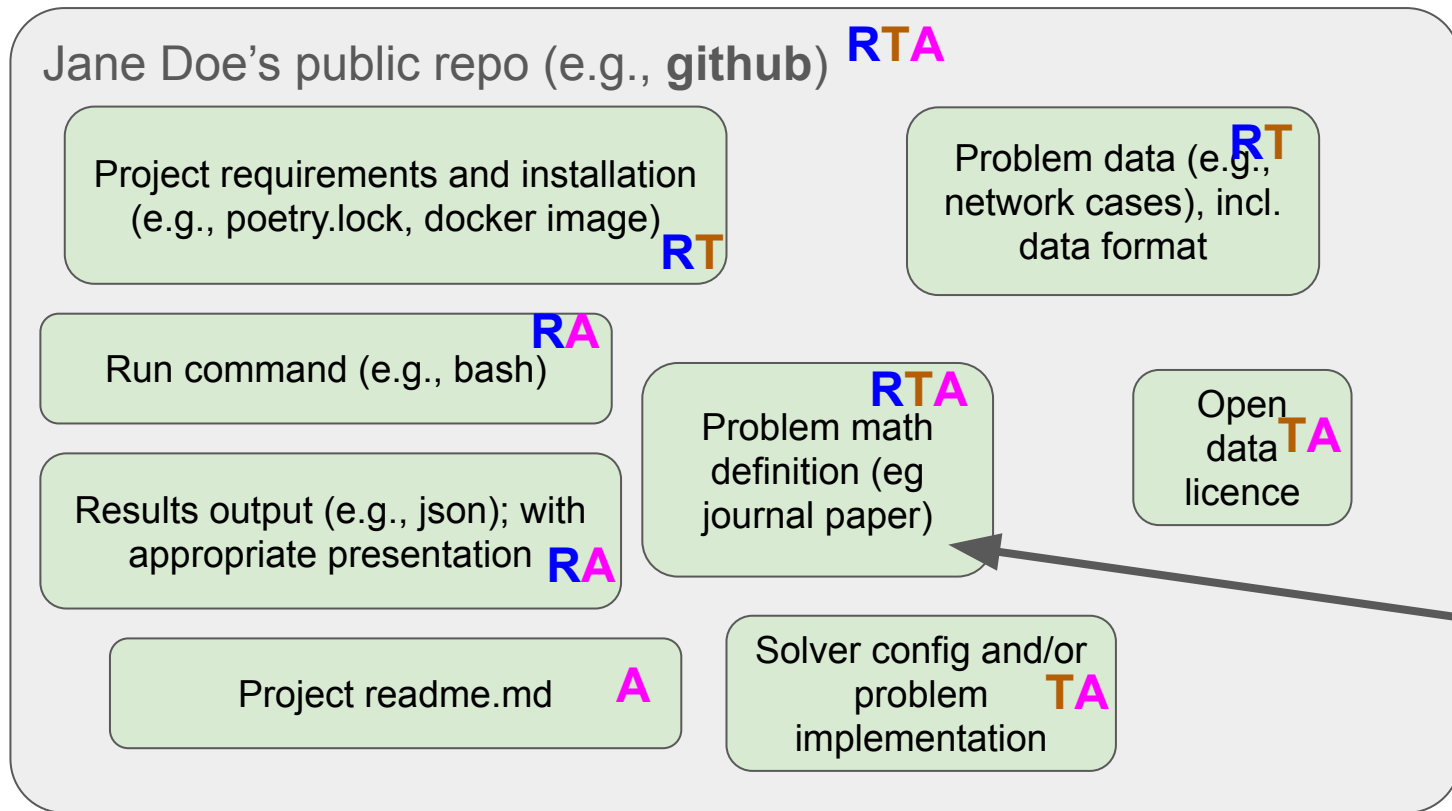
[Link to data spec](#)

Data Specification: some deliberate choices

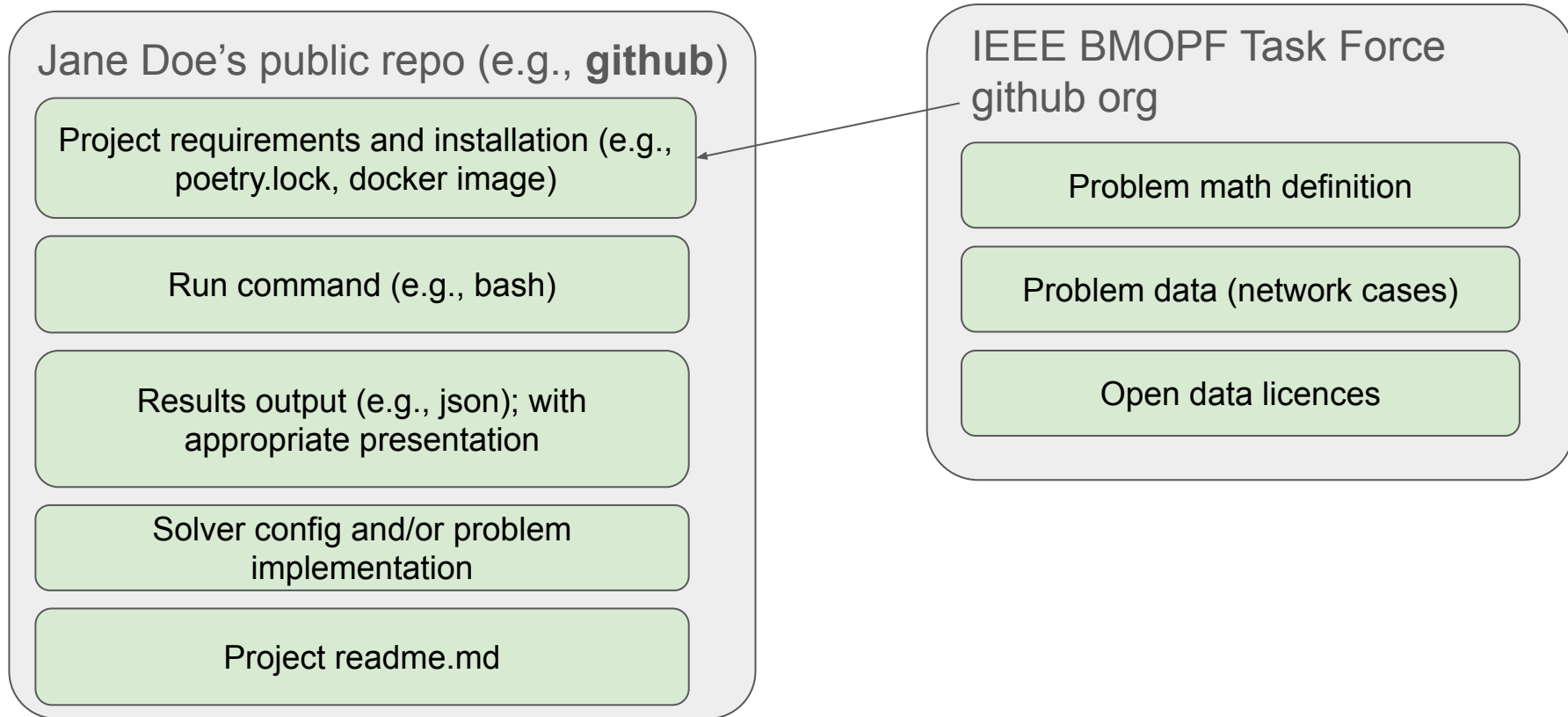
- The use of “wire coordinates”
- Nonlinear current-voltage, rectangular (“IVR”) formulation
- Exclusively SI units for physical quantities (no per-unit transformer impedances)
- Support meshed networks & parallel branches
- Grounding for all components made explicit as rules

...read the full spec for more!

Towards A Distribution System OPF Toolkit: what is necessary for **R**eproducible, **T**ransparent and **A**ccessible Research



Solving benchmarks defined by the TF enables substantially reduced overhead for Transparent, Reproducible and Accessible Research



A (Work-in-Progress) **Distribution System OPF Toolkit** to enable Reproducible, Transparent and Accessible outputs

Core BMOPF usage:

- Git (+ github / gitlab etc). (Many tutorials.)
- BMOPF Data Specification - the basic network math, physics, json format
- BMOPF Network Cases - OPF-ready network cases + validated results

Helpers (all community tools)

- 3 PowerUp Case Studies
- Moving to + from network formats - PowerIO
- Building & inspection of OPF cases - [BMOPFTools.jl](#)
- Solvers of BMOPF (ACOPF) problems - [BMOPFTools.jl](#), tellegen.dev
- Vis - [DistributionNetworkPlots.jl](#); tellegen.dev

What about problems that are not benchmarks?

Route 1: create new benchmark problem specification

- **Open an Issue** on the BMOPF data specification site discussing the new problem (new components, new objective function, etc)
- Develop proposal to update data spec according to guidelines (**contributing.md**)
 - Updated JSON Schema
 - Update corresponding math
 - Develop / adapt network cases
- **Open “Pull Request”**; if successful, becomes new benchmark problem.

-> More time consuming but community review ensures accessibility etc

Route 2: use BMOPF infrastructure as basis

- Use BMOPF physics for network; use network cases as basis
- Using “extras” in JSON for extra data required to augment the problem

-> Misses out on ‘best known solution’ aspect of benchmark problems

Tool demo and interactive case studies

PowerIO:
Compiler
Infrastructure for
Power Systems
(S. Talkington)

Q: how can I get
my .dss file into
BMOPF json?

From High-Quality
Data Models to
High-Fidelity
Network Models
(M. Vanin)

Q: Why
wire-coordinates
as the level of
abstraction?

Modeling Smart
Inverters in
Distribution OPF
(A. Inaolaji)

Q: why if-else
conditions have to
be modelled
systematically in
OPF?

Physics-based
OPF with IBRs in
Distribution Grids
(A. Pandey)

Q: how can the
BMOPF data
specification be
extended for
highly granular
OPF problem?

PowerIO: Compiler Infrastructure for Power Systems

Sam Talkington

[Link to slides](#)

Interactive Case Study 1

From High-Quality Data Models to High-Fidelity Network Models

Marta Vanin

[Link to notebook](#)

Break 1 - 10 mins

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How is Distribution OPF different from Transmission OPF?

Amrit Pandey, Marta Vanin

(Incomplete) Taxonomy of transmission OPF

Capacity expansion problem (CEP)	Simple network model (linearized)	Multi-period	MILP
Production cost model (PCM)	Simple network model (linearized)	Multi-period	MILP
Security-constrained unit commitment (SCUC)	Simple network model (linearized)	Multi-period	MILP
Stochastic unit commitment	Simple network model (linearized)	Multi-period	MILP
Optimal Power Flow (OPF)	AC network model	Single-period	NLP
Multi-period OPF (MP-OPF)	AC or relaxed network model	Multi-period	NLP
Security constrained ACOPF (SC-OPF)	AC network model	Single period	NLP
Stochastic ACOPF	AC network model	Single period	NLP
Transmission expansion problem (TEP)	AC network model	Multi-period	MINLP

Pandey, Almassalkhi, & Chevalier, "Large-scale grid optimization: the workhorse of future grid computations," Curr. Sustain. Renew. Energy Rep., 10, 139-153 (2023).

What is solved and what is not solved in transmission OPF?

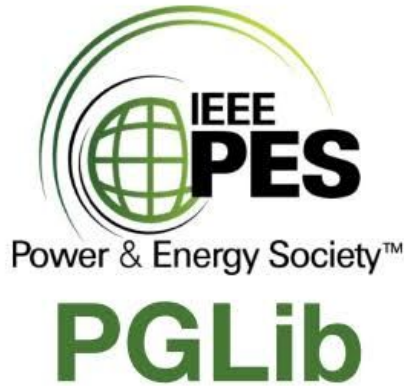
Solved

- Formulations are well understood
- Relaxations are well studied and documented
- Competitions and existing data have illustrated the trade-offs well
- Predominantly driven by academics but increasingly used in the real world

Not solved

- Benchmarking
- Real challenge in my opinion: 100s of papers on OPF each year, yet there is no consensus on the “winning” approach

State-of-the-art in Transmission OPF Datasets (Afaik)



- De-facto benchmark
 - Includes baselines
 - **Not live - Last commit circa 2023 :-)**
 - Baselines outdated



- ARPA-E Grid Optimization
 - Industry-partnered
 - **Only metadata available at**
data.opennei.org

Frontier now in ML-based learning-to-optimize datasets

OPFData: Large-scale datasets for AC optimal power flow with topological perturbations

Sean Lovett^{1,*}, Miha Zgubič^{1,*}, Sofia Liguori¹, Sephora Madjiheurem¹, Hamish Tomlinson¹, Sophie Elster¹, Chris Apps¹, Sims Witherspoon¹ and Luis Piloto¹

¹Google DeepMind, *Equal contribution

Maybe largest public collection: around 13,000 solved AC-OPF instances across 8 grid models ranging from 118 to 6,515 buses!

I say, maybe, too little!

Benchmarking strategy - Reproducibility and Transparency

- Pin down the instance
 - Case name, ...
- Pin down the problem
 - Formulation, ...
- Pin down the terminal condition
 - Residual at convergence, ...
- Pin down the timing and iteration count
 - Initial state, ...
- Release a one-command repo
- Keep it live and community driven!

Examples of Successful Benchmarks



Open LLM Leaderboard Archived

Comparing Large Language Models in an open and reproducible way

ImageNet.

ImageNet Large Scale Visual Recognition Challenge (ILSVRC): the standard 1,000-class image classification benchmark. Sparked the deep learning revolution from 2010 onward.

Benchmarking a non-convex vs convex NLP routine

Non-convex exact:

- Feasible objective
- Contextualize with lowest found feasible objective (current UB)

Convex relaxation:

- Dual bound
- Contextualize gap with current best UB

Approximate:

- It's a different optimization problem altogether

T vs D

	Transmission	Distribution
Model specifications	Mature	Inconsistent
Benchmarks	Could be better	Non-existent
OPF Formulation	Mature	Highly variable
OPF Solver	Mature	No industrial or academic consensus on best approaches
Industry has good data	Yes	No

T vs D

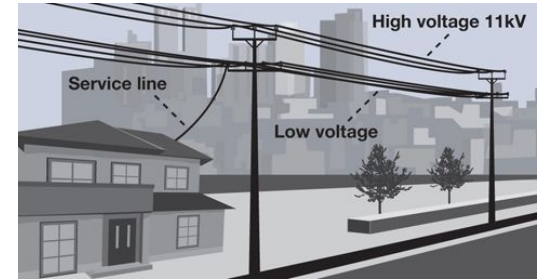
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Discussed before in the workshop + reasons why this TF exists

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Distribution OPF formulation



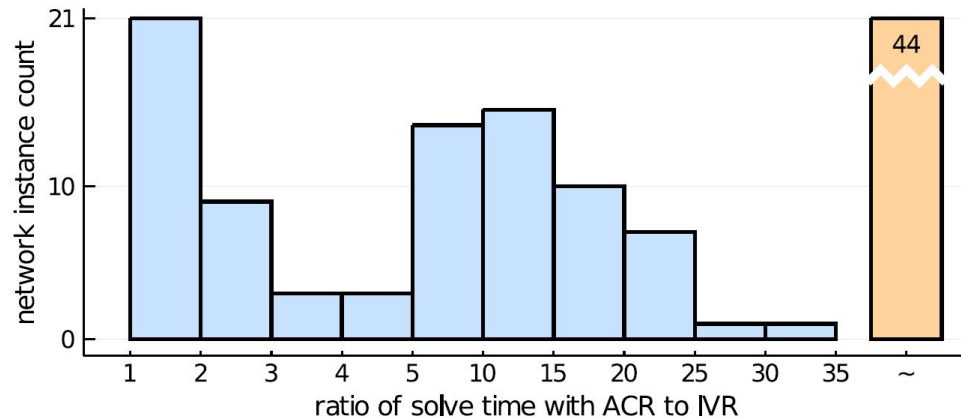
Industry needs to solve them all, including occasional meshed network, explicit four-wire, swer, ...

Hence the power of “wire coordinates”

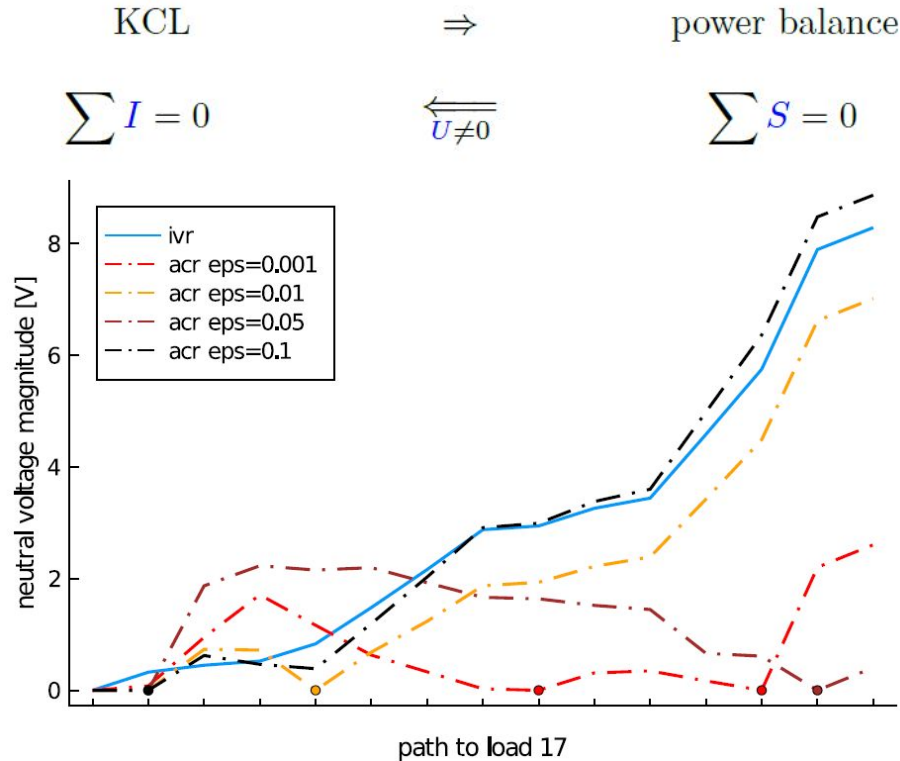
Padding conductors and omitting service lines still rather common

Distribution OPF formulation - why the IVR formulation

$$\begin{array}{ccc} \text{KCL} & \Rightarrow & \text{power balance} \\ \sum I = 0 & & \sum S = 0 \end{array}$$



Distribution OPF formulation - why the IVR formulation



Despite different eps+j0 initialization of the neutral voltage variables lead to ACR objectives [-2.78; -2.75].

IVR not affected

Distribution OPF formulation - why the IVR formulation

KCL

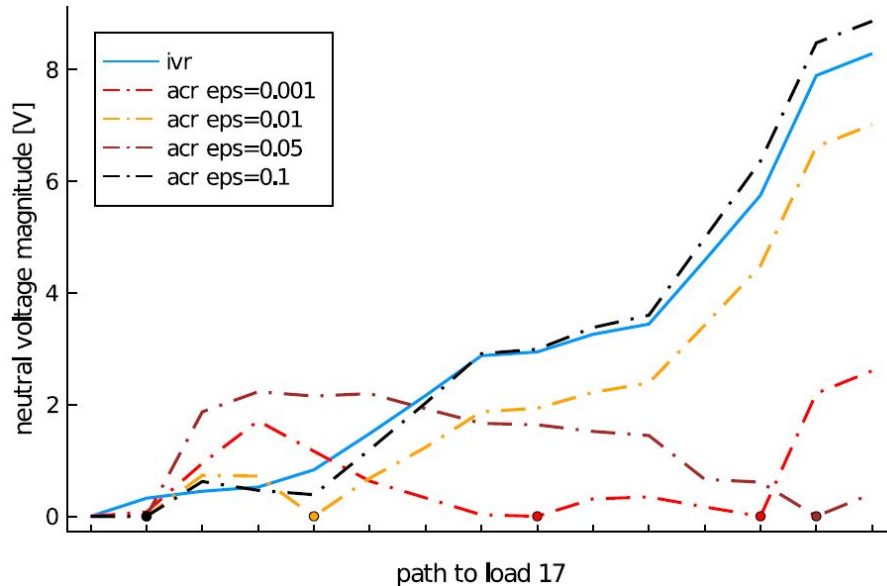
$\Rightarrow$

power balance

$$\sum I = 0$$

$$\overleftarrow{U} \neq 0$$

$$\sum S = 0$$



ACR's incorrect solutions = grounding the neutral at one or many buses.

Typically this improves the network transfer capacity

Pitfalls of Zero Voltage Values in Optimal Power Flow Problems

Frederik Geth

<https://ieeexplore.ieee.org/abstract/document/10253442>

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Distribution OPF (and related problems) solvers

OPF: model and solvers decoupled, solver agnostic → thanks to automatic diff. toolchains!

- e.g., (PowerModelsDistribution/BMOPFTools+)JuMP+Ipopt
- Probably standard in the research community (anyone still implements own solvers?)
- License bottlenecks for (some) companies(?)
- Just not a lot of “OPF proper” in ADMS

Power flow solvers

- BFSweep still a thing (at least until few years back) → does not solve meshed networks
- Legacy solvers from older projects / implementations → harder than building from scratch
- Could be implemented like an OPF (and is, e.g., in PMD/BMOPFTools) but native solvers often >> faster
- Not interesting for research anymore(?)
- ADMS and similar commercial products should have good enough solutions

Distribution OPF (and related problems) solvers

Distribution system state estimation

- Both interesting and challenging from research and industry perspective
- Unconstrained (transmission-like WLS) algorithms still dominating; lots of new papers use ML
 - Neither of them solid for distribution at all times (right now)
- Very odd “state estimators” in common ADMS
 - Single-phase eq. reduction; no low voltage support, “load allocation” sold as “state estimation”

General issues with ADMS

- Not very innovative: few large vendors vs many small utilities, lots of inertia vs limited business cases
 - (Very) few digital measurements or controllable assets until quite recently
- ...

Distribution OPF (and related problems) solvers

No longer an issue with GenAI/agents?



Tom Brown ✓ • 2°
Professor of Energy Systems at TU Berlin; Managing Director PyPSA ...
1h • 



Agentic AI is surely the Killer App of open energy modelling - just watching the agents play with **PyPSA** workflows, it's hard to imagine how closed suites can compete.

Just the sheer amount of online code and examples with **PyPSA** that the agents are trained on means they're super-familiar with the tooling, concepts and conventions of open source tools. And when they do get stuck, they can go and look it up, search the issue tracker and read or even edit the actual source code. Closed source gives them much less to work with.

With relatively little prompting they can go and do sophisticated things that would have taken days in the past. With future agent tooling, this will only improve.

(...) read Tom Brown's full post:

<https://www.linkedin.com/feed/update/urn:li:activity:7501274070164930560/>

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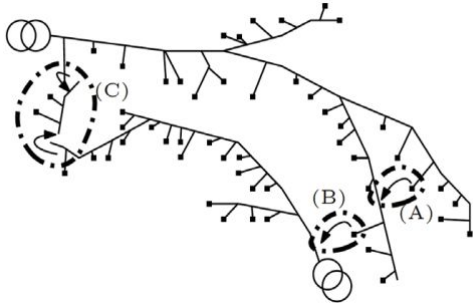
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Is utilities' slow digitalization and lack of trustworthy network data the killer of GenAI's advantages? Can the taskforce models help?

Distribution industry does not have good data

Table 1 Data issues, an asterisk indicates particularly under-addressed ones.

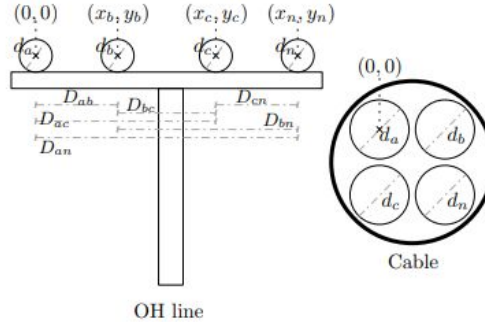
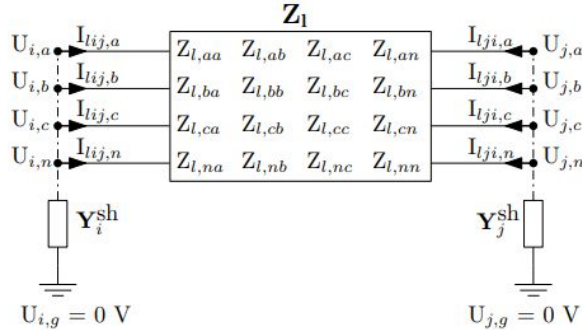
	Issue	Implications and risks
1	Missing phase labels (loads, transformers, etc.)	Inaccurate per-phase power flows, voltage unbalance inexact.
2	Switch states & user-branch connections	See issue 1 + impossible to determine how many consumers are connected to a transformer (aggregate values at feeder head inaccurate).
3*	Meter-to-transformer assignment	See issue 2.
4*	Excessive number of buses	Errors if reductions are performed, increased computational time.
5	Wrong transformer tap	Inaccurate estimates of: end-consumer voltages, hosting capacity, voltage-based curtailment.
6	Missing tap semantics	Taps are usually given integer values, the nominal value is generally not 0, the tap percentage can be unknown, see issue 5.
7*	Unknown vector group usage (ansi vs euro)	Sign error in angle off-set between primary and secondary for Dy transformers.
8*	Unknown winding configuration	Inaccurate voltage values, problematic for harmonic studies.
9*	Mislabeled transformer primary/secondary nominal voltage	See issue 5 + this also implies wrong transformer impedance (as they are typically specified in per unit w.r.t. the transformer power rating and primary voltage).
10*	Wrong transformer rating	Inaccurate classification of congestion; also implies inaccurate transformer impedance (as they are typically specified in per unit w.r.t. the transformer power rating and primary voltage).
11*	Only Kron-reduced impedance matrices available	Assumes neutral voltage rise is marginal under normal conditions, which is likely problematic in sparsely grounded networks. Not compatible with short circuit analysis.
12*	Only sequence impedances available	Equivalent to assuming transposition + multi-grounding of neutral in 4-wire networks.
13*	Meter-phase-alignment for three-phase users	Inaccurate SE and other measurement-based computations.
14*	Missing information on neutral grounding	Inaccurate voltage and current estimates, impact depends on grounding philosophy.
15	Load model (constant power, ZIP) unknown	Analyses may be inaccurate in terms of voltages, unbalance levels, neutral current.
16	Missing load (locations)	SE may increase load at the known locations, leading to inaccurate congestion identification.
17	Measurements rms vs fundamental-only	When using a fundamental-frequency-only (O)PF or SE, contributions of the higher frequencies to the RMS values may lead to inaccuracy.
18*	Regulators modelled as transformers	Impedance values differ between transformers and regulators, inaccurate voltages/currents.
19	Missing capacitor banks (specifications)	Reactive power flows / power factor for loads/generators look statistically unlikely.
20	Approximate cable/line impedance models	Inaccurate estimation of currents (particularly neutral current), and voltage drops.
21*	Unknown (PV) inverter settings	Constant power factor and volt-var/watt lead to very different patterns of overvoltage.
22*	Unknown home battery dispatch strategies	Complementarity between PV, batteries & load overestimated, inaccurate voltages/currents.



F. Geth, M. Vanin, D. Van Hertem, “Data quality challenges in existing distribution network datasets”, 27th International Conference on Electricity Distribution, Rome, 12– 15 June 2023

Distribution industry does not have good data

With a good data model we can learn/fit something which matches the physics, from the sparsely available measurements. Work in progress..



$$Z_{pp,k}^{\text{nom}} = R_{pp,k}^{\text{nom}} + jX_{pp,k}^{\text{nom}} \quad \forall k \in \mathcal{K}, \quad \left[\frac{\Omega}{\text{km}} \right]$$

$$Z_{pq,k}^{\text{nom}} = R_{pq,k}^{\text{nom}} + jX_{pq,k}^{\text{nom}} \quad \forall k \in \mathcal{K}, \quad \left[\frac{\Omega}{\text{km}} \right]$$

$$R_{pp,k}^{\text{nom}} = r_{p,k} + 0.049348,$$

$$R_{pq,k}^{\text{nom}} = 0.049348,$$

$$X_{pp,k}^{\text{nom}} = 0.062832 \left(\ln \left(\frac{1}{c_1 f_{p,k}^{\text{GMR}}} \right) + c_2 \right),$$

$$X_{pq,k}^{\text{nom}} = 0.062832 \left(\ln \left(\frac{1}{c_1 D_{pq,k}} \right) + c_2 \right).$$

Interactive Case Study 2

Modeling Smart Inverters in Distribution OPF

Adedoyin Inaolaji

[Link to slides](#)

Break 2 - 10 mins

1. How can the distribution system OPF community be encouraged to work on industrially relevant problems?
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Interactive Case Study 3

Physics-based OPF with IBRs in Unbalanced Distribution Grids

Amritanshu Pandey

[Link to slides](#)

The Future of Distribution System OPF



[Join mailing list](#)

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Round-table reflective discussion

Facilitated discussions

Thanks for your time!

Towards a New Era of Reproducible and Accessible Distribution System OPF Research

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Sponsor: IEEE Task Force on Benchmarking Multiconductor OPF for Distribution Systems

